Chapter 1: The Living World

Biology is the science that studies life forms and living processes. The living world has an amazing diversity of organisms.

1.1 Diversity in the Living World

- The living world shows a wide range of living types and extraordinary habitats where organisms are found, like cold mountains, oceans, and hot springs.
- A key question in biology is "what is living?" as opposed to non-living.
- If you observe a wider area, the variety of organisms you see increases.
- Each different kind of plant, animal, or organism represents a

  species.

- The number of known and described species is between 1.7-1.8 million. This is referred to as

  biodiversity. New organisms are continuously being identified.

- **Nomenclature** is the process of standardizing the naming of living organisms so that a particular organism is known by the same name worldwide.
- **Identification** is necessary for nomenclature; it's when an organism is correctly described, and we know which organism the name refers to.
- Scientific names for plants are based on the

  International Code for Botanical Nomenclature (ICBN).

- Animal taxonomists have developed the

  International Code of Zoological Nomenclature (ICZN) for naming animals.

- Scientific names ensure that each organism has only one name and that this name hasn't been used for any other known organism.
- **Binomial nomenclature** is the system of providing a scientific name with two components: the Generic name and the specific epithet. This system was given by Carolus Linnaeus.
- **Universal rules of nomenclature:**
    - Biological names are generally in Latin and written in italics. They are Latinized irrespective of their origin.
    - The first word is the genus, and the second component is the specific epithet.
    - Both words are separately underlined when handwritten or printed in italics to indicate their Latin origin.
    - The genus name starts with a capital letter, and the specific epithet starts with a small letter (e.g.,

      Mangifera indica).

- The author's name appears after the specific epithet in an abbreviated form (e.g.,

  Mangifera indica Linn.), indicating who first described the species.

- **Classification** is the process of grouping organisms into convenient categories based on easily observable characteristics.
- **Taxa** (singular: taxon) are the scientific terms for these categories (e.g., 'Dogs', 'Cats', 'Mammals', 'Wheat', 'Plants', 'Animals'). Taxa can represent categories at different levels (e.g., 'animals', 'mammals', 'dogs' are all taxa at different levels).
- **Taxonomy** is the process of classifying living organisms based on their characteristics. Modern taxonomic studies rely on external and internal structure, cell structure, development process, and ecological information.
- The basic processes of taxonomy are characterization, identification, classification, and nomenclature.
- **Systematics** is the branch of study that deals with the diversity of organisms and their relationships. The word 'systematics' comes from the Latin word 'systema', meaning systematic arrangement of organisms. Linnaeus used "Systema Naturae" as a publication title. Systematics now includes identification, nomenclature, classification, and evolutionary relationships.

1.2 Taxonomic Categories

- Classification is a hierarchy of steps, where each step is a rank or category. These categories together form the

  taxonomic hierarchy.

- Each category is called a **taxon** (plural: taxa).
- Common taxonomic categories from highest to lowest are: Kingdom, Phylum (or Division for plants), Class, Order, Family, Genus, and Species. Species is the lowest category.
- To place an organism into categories, knowledge of its characters is essential to identify similarities and dissimilarities.

1.2.1 Species

- A **species** is a group of individual organisms with fundamental similarities.
- Different species can be distinguished based on distinct morphological differences.
- Examples: *Mangifera indica* (mango), *Solanum tuberosum* (potato), and *Panthera leo* (lion). The words

  indica, *tuberosum*, and *leo* are specific epithets.

- Human beings belong to the species *sapiens* in the genus *Homo*, thus *Homo sapiens*.

1.2.2 Genus

- A **genus** comprises a group of related species with more common characters compared to species of other genera. Genera are aggregates of closely related species.
- Example: Potato and brinjal are different species but both belong to the genus

  Solanum.

- Lion (*Panthera leo*), leopard (*P. pardus*), and tiger (*P. tigris*) are all species of the genus *Panthera*. This genus is different from

  Felis (cats).

1.2.3 Family

- A **family** is a group of related genera with fewer similarities than within a genus or species.
- Families are characterized by both vegetative and reproductive features (for plants).
- Examples:
  - Plant genera

    Solanum, *Petunia*, and *Datura* are placed in the family Solanaceae.

  - Animal genera

    Panthera (lion, tiger, leopard) and *Felis* (cats) are in the family Felidae.

  - Cats and dogs belong to different families: Felidae and Canidae, respectively.

1.2.4 Order

- **Order** is a higher category, an assemblage of families that show a few similar characters. The similar characters are less numerous than within a family.
- Examples:
  - Plant families Convolvulaceae and Solanaceae are in the order Polymoniales, mainly based on floral characters.
  - The animal order Carnivora includes families like Felidae and Canidae.

1.2.5 Class

- A **class** includes related orders.
- Example: Order Primata (monkey, gorilla, gibbon) and order Carnivora (tiger, cat, dog) are both placed in class Mammalia.

1.2.6 Phylum

- **Phylum** is the next higher category above Class. Classes like fishes, amphibians, reptiles, birds, and mammals constitute a phylum.
- Organisms with common features like the presence of notochord and dorsal hollow neural system are included in Phylum Chordata.
- In plants, classes with similar characters are assigned to a higher category called **Division**.

1.2.7 Kingdom

- **Kingdom** is the highest category.
- All animals belonging to various phyla are assigned to Kingdom Animalia.
- Kingdom Plantae comprises all plants from various divisions.
- As we go higher from species to kingdom, the number of common characteristics among members decreases.
- Lower the taxa, more characteristics are shared by its members. Higher the category, the more difficult it is to determine relationships to other taxa at the same level.